

RepoQuest

Turn an unfamiliar repo into an onboarding quest.

A Streamlit app that scans a small codebase, maps its architecture, creates a reading path, builds component cards, quizzes new contributors, and exports a Markdown guide.

Presented by

Ovidiu-Ştefan Andreescu | Sirbu-Boeti-Eduard

safe static analysis

AI Fusion optional

demo-ready

Scan repo

Map architecture

Create quest

Static analysis only. No default secrets required.

The first 30 minutes vanish into orientation work.

A new contributor has to infer the project type, entry points, routes, data models, tests, and architecture before making a useful change.

01

Where do I start?

Random file browsing

02

What kind of repo is this?

Framework clues scattered across config and code

03

What talks to what?

Imports, routes, services, and tests are disconnected

04

What should I read first?

No contributor path or defense questions

COST

Onboarding debt repeats.

Every new contributor asks about entry points, framework, architecture, tests, and next action. Generic chat can help, but it can also hallucinate or require setup.

Orientation debt loop

Search

Guess

Ask again

The same orientation work repeats until the repo has a map.

RepoQuest turns scattered repo structure into contributor-ready artifacts.

The app analyzes a bundled demo repo or uploaded ZIP, then outputs the concrete objects a developer needs to onboard.



Deterministic evidence comes first; interpretation comes after.

RepoQuest does not start with a chatbot. It builds a transparent evidence pack from static analysis, then turns it into onboarding artifacts.



Hard facts stay authoritative: scanned files, skipped files, routes, imports, graph data, and ZIP safety warnings.

Optional AI Fusion can improve summaries and rankings only after citing scanned evidence.

Safety is built into the scanner, not bolted on after upload.

The default demo requires no credentials. Uploaded ZIPs are scanned defensively and never executed.

Protection	What RepoQuest does	Why it matters
 No execution	Never runs uploaded scripts or imports uploaded modules	safe by default
 No install	Never installs dependencies from the uploaded repo	safe by default
 ZIP slip checks	Rejects absolute paths and path traversal attempts	bad files do not crash the flow
 Limits	25 MB ZIP, 600 files, 512 KB per file, 80 graph nodes	demo stays fast
 Binary skip	Ignores binary and unsupported formats	bad files do not crash the flow

The live demo shows the full value in under two minutes.

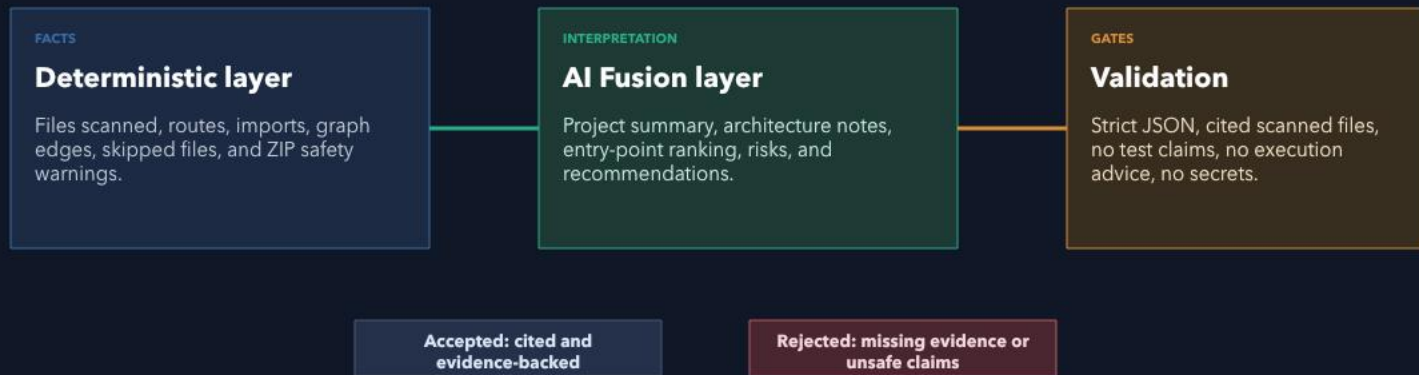
Use the bundled Mini Travel Planner repo: a small React + Vite and FastAPI app with frontend, backend, routes, services, models, and tests.



Judges see React/Vite + FastAPI detection, route extraction, graph rendering, reading path, component cards, quiz, and export without connecting GitHub.

AI can improve interpretation, but it cannot rewrite scanner facts.

RepoQuest sends a bounded evidence pack to AI only when configured, then validates the structured response before showing it.



RepoQuest turns discovery into a reusable artifact, not a one-off conversation.

The same output supports four audiences without changing the core workflow.

AUDIENCE 1

Developer

Knows where to start and what to read first.

AUDIENCE 2

Reviewer

Gets repo context before judging or triaging.

AUDIENCE 3

Team

Receives onboarding docs and defense questions.

AUDIENCE 4

AI assistant

Gets better prompts grounded in real files.

Output: fingerprint + architecture + reading path + cards + quiz + Markdown guide + Bob-ready prompts

IBM Bob accelerated the build; RepoQuest stays independently deployable.

Bob helped create the app, tests, docs, and iteration trail. The app itself runs on Streamlit without IBM Bob at runtime.



Default mode remains deterministic, safe, fast, and credential-free. Optional AI Fusion is configured separately and validation-gated.

RepoQuest helps developers answer the real first question: where do I start?

A safe static analyzer, a practical onboarding guide, and an optional AI interpretation layer for teams that want deeper synthesis.

- ✓ **Deterministic – same repo, same evidence**
- ✓ **Safe – no uploaded code execution**
- ✓ **Fast – demoable in 1-2 minutes**
- ✓ **Deployable – Streamlit Cloud friendly**
- ✓ **Actionable – exports prompts and guide**

Presentation

RepoQuest

Turn an unfamiliar repo into an onboarding quest.

IBM Bob Hackathon

Ovidiu-Ștefan Andreescu
Sirbu-Boeti-Eduard